

Table S1a: Model GWL Crossing Years and Storyline Assignments (SSP5-8.5)							
Model	Scenario	GWL +2C Year	Winter SL (+2C)	Summer SL (+2C)	GWL +3C Year	Winter SL (+3C)	Summer SL (+3C)
ACCESS-CM2	SSP5-8.5	2039	Increasing Frequency	Increasing Frequency	2056	Decreasing Frequency	Neutral / Other
ACCESS-ESM1-5	SSP5-8.5	2040	Increasing Frequency	Increasing Frequency	2061	Increasing Frequency	Increasing Frequency
BCC-CSM2-MR	SSP5-8.5	2044	Increasing Frequency	Increasing Frequency	2066	Increasing Frequency	Increasing Frequency
CESM2	SSP5-8.5	2037	Increasing Frequency	Increasing Frequency	2056	Increasing Frequency	Increasing Frequency
CESM2-WACCM	SSP5-8.5	2034	Neutral / Other	Neutral / Other	2054	Increasing Frequency	Increasing Frequency
CMCC-CM2-SR5	SSP5-8.5	2034	Increasing Frequency	Increasing Frequency	2053	Decreasing Frequency	Increasing Frequency
CMCC-ESM2	SSP5-8.5	2040	Increasing Frequency	Increasing Frequency	2056	Increasing Frequency	Increasing Frequency
CNRM-CM6-1	SSP5-8.5	2041	Increasing Frequency	Increasing Frequency	2059	Increasing Frequency	Increasing Frequency
CNRM-CM6-1-HR	SSP5-8.5	2030	Increasing Frequency	Increasing Frequency	2052	Neutral / Other	Neutral / Other
CNRM-ESM2-1	SSP5-8.5	2046	Decreasing Frequency	Neutral / Other	2065	Decreasing Frequency	Decreasing Frequency
CanESM5	SSP5-8.5	2023	Decreasing Frequency	Decreasing Frequency	2041	Decreasing Frequency	Decreasing Frequency
E3SM-1-0	SSP5-8.5	2033	Decreasing Frequency	Decreasing Frequency	2048	Decreasing Frequency	Decreasing Frequency
EC-Earth3	SSP5-8.5	2036	Decreasing Frequency	Decreasing Frequency	2058	Decreasing Frequency	Decreasing Frequency
EC-Earth3-CC	SSP5-8.5	2037	Decreasing Frequency	Neutral / Other	2057	Neutral / Other	Decreasing Frequency
EC-Earth3-Veg-LR	SSP5-8.5	2042	Decreasing Frequency	Decreasing Frequency	2062	Decreasing Frequency	Decreasing Frequency
GFDL-ESM4	SSP5-8.5	2053	Decreasing Frequency	Decreasing Frequency	2076	Decreasing Frequency	Decreasing Frequency
HadGEM3-GC31-LL	SSP5-8.5	2031	Increasing Frequency	Decreasing Frequency	2048	Increasing Frequency	Decreasing Frequency
HadGEM3-GC31-MM	SSP5-8.5	2035	Increasing Frequency	Increasing Frequency	2050	Increasing Frequency	Neutral / Other
IITM-ESM	SSP5-8.5	2049	Neutral / Other	Decreasing Frequency	2077	Decreasing Frequency	Decreasing Frequency
INM-CM4-8	SSP5-8.5	2047	Decreasing Frequency	Decreasing Frequency	2070	Increasing Frequency	Increasing Frequency
INM-CM5-0	SSP5-8.5	2046	Increasing Frequency	Increasing Frequency	2075	Neutral / Other	Increasing Frequency
IPSL-CM6A-LR	SSP5-8.5	2035	Decreasing Frequency	Decreasing Frequency	2051	Decreasing Frequency	Decreasing Frequency
KACE-1-0-G	SSP5-8.5	2024	Neutral / Other	Increasing Frequency	2044	Increasing Frequency	Increasing Frequency
KIOST-ESM	SSP5-8.5	2039	Increasing Frequency	Increasing Frequency	2065	Increasing Frequency	Increasing Frequency
MIROC-ES2L	SSP5-8.5	2048	Decreasing Frequency	Decreasing Frequency	2071	Decreasing Frequency	Decreasing Frequency
MIROC6	SSP5-8.5	2054	Decreasing Frequency	Decreasing Frequency	2077	Decreasing Frequency	Decreasing Frequency
MPI-ESM1-2-HR	SSP5-8.5	2050	Increasing Frequency	Increasing Frequency	2074	Increasing Frequency	Increasing Frequency
MPI-ESM1-2-LR	SSP5-8.5	2049	Decreasing Frequency	Decreasing Frequency	2072	Increasing Frequency	Increasing Frequency

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Model	Scenario	GWL +2C Year	Winter SL (+2C)	Summer SL (+2C)	GWL +3C Year	Winter SL (+3C)	Summer SL (+3C)
MRI-ESM2-0	SSP5-8.5	2039	Decreasing Frequency	Decreasing Frequency	2065	Decreasing Frequency	Decreasing Frequency
NorESM2-MM	SSP5-8.5	2055	Increasing Frequency	Increasing Frequency	2077	Increasing Frequency	Increasing Frequency
UKESM1-0-LL	SSP5-8.5	2032	Decreasing Frequency	Decreasing Frequency	2047	Decreasing Frequency	Decreasing Frequency